Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Seattle, WA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) researching agentic LLM systems, RAG pipelines, and MCP tool-calling infrastructure. Production engineering background building distributed backend systems and cloud automation using Python, Java, AWS, and Kubernetes.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Deep Learning, Intelligent Data Analysis, Advanced Algorithms.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Machine Learning, Data Mining, Data Analytics.

TECHNICAL SKILLS

Programming: Python, Java, TypeScript, JavaScript, Go, SQL, C/C++.

AI & LLM: Gemini, Vertex AI, RAG, MCP (Model Context Protocol), LangChain, PyTorch, TensorFlow, scikit-learn, Prompt Engineering, LLM Evaluation.

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices Architecture, Zod, WebSockets.

Data: PostgreSQL, Vector Databases, Redis, DynamoDB, pandas, NumPy.

Cloud & DevOps: AWS, Docker, Kubernetes, GitLab CI/CD, Terraform, Linux, Git.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented **LLM pipeline** decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built provider-agnostic LLM abstraction layer (Gemini, GPT-4o, Llama, Qwen, MiniCPM), isolating method effects from model choice; demonstrated **25.7x latency reduction** while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault; designed Ansible-based Blue-Green deployment reducing downtime **50%**.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.
- Engineered confidence-scoring system (evidence count, negation detection) driving human-review triage, migrating from Ollama to Gemini after benchmarking hallucination rates.

Spendora – Conversational AI for Expense Insights [LangChain, HuggingFace, Mistral, Gemma] Jun 2025

- Developed LLM-powered finance assistant with **RAG** and custom toolchains, deploying conversational agents with memory and tool use to automate expense categorization and surface spending patterns.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).